DESIGN SYSTEMS

TECHNICAL DATA SHEET DS-POLTEC SOL A+B

SOLVENT-BASED TRANSPARENT POLYURETHANE VARNISH – GLOSSY, SATIN AND MATT

DESCRIPTION

DS-POLTEC SOL A+B is a two-component, solvent-based, transparent polyurethane varnish, formulated with aliphatic polyurethane resins and available in glossy, satin and matt versions. Suitable for finishing resin coatings or concrete floors, with excellent mechanical and chemical resistance.

- Forms a film that maintains good flexibility and good resistance to abrasion and scratching.
- Resists ultraviolet rays thanks to UV filters that prevent its photo-oxidative degradation.
- The cured film resists attack by many acids.
- Excellent mechanical and chemical resistance.

USES

Finish for resin coatings. Varnishing of concrete floors. Anti-corrosion finishing treatment for concrete and iron structures. Particularly suitable for battery storage areas.

SUBSTRATE PREPARATION

Every substrate must be clean, degreased and free from old coatings that are flaking off. Lightly abrade the surface to promote adhesion. It is advisable to use light-coloured abrasive papers with grit 150 or finer. After abrading, remove all dust from the surface completely.

- Existing resin coatings must be abraded or ground, and the dust residue then removed. In the case of newly laid resin coatings, the overcoating time of the previously applied product must be checked.
- Concrete and brick surfaces must be sound, cured, level, absorbent and not contaminated by oils, dust or other substances. Prepare the surface mechanically by sanding or grinding and prime with a suitable primer.
- Metal surfaces: apply an anti-corrosion epoxy primer after mechanical preparation.

APPLICATION

Combine the two components (A+B) and mix using a mixing drill for 1 minute. Apply the mixture by roller with a consumption of approximately 0,100 kg/m^2. To obtain a rough, non-slip surface, add 3-5% of GLASS BEADS or 10-20% of QUARTZ B0/B1 when preparing the two components (see technical data). Avoid preparing partial mixes of the product so as not to incur accidental errors that could result in failure to cure or incomplete curing. If a second coat is applied, sand beforehand.

If component A has to be decanted from its original container, mix carefully in order to homogenise any matting agents that may have settled and thus obtain the nominal gloss value.

WARNINGS

Owing to the nature of this type of product, pay particular attention during application in order to avoid defects such as roller marks, overlaps, differences in gloss, etc. In general, spread the varnish evenly over the substrate in the right quantity; when joining up with the area just varnished, resume varnishing a few centimetres away so as to discharge the excess product and then blend into the previous area, feathering and carefully dosing the quantity applied. Do not wait too long between overlaps: always varnish "wet-on-wet". When covering large surfaces, the coordinated movement of several applicators is required. Brush application produces a film with a higher final gloss than roller application.

The product contains solvents: applicators must wear the appropriate respiratory protective equipment indicated in the safety data sheet. It is recommended to suitably compartmentalise the work areas during application and in the following hours. During the evaporation phase, the nature of the solvents used may create unpleasant odours for persons not involved in the work who are present in nearby areas: whenever possible, ventilate the rooms well.

Product for professional use: the purchaser undertakes to strictly follow the warnings given above when applying the purchased product, as well as the instructions in the safety data sheet.

PRODUCT DATA

DESCRIPTION	RESULT
Colour	Transparent
Consumption	By roller: approx. 0,100 kg/m^2
Specific gravity (at 25°C)	Mixture A+B: 1,00 ± 0,05 g/ml
Viscosity (at 25°C)	Mixture A+B: 500 ± 50 mPa·s (spindle 2, 30 rpm)
Solids content (A+B)	59-61% by weight
VOC ready for use (Italian Legislative Decree 161/06)	< 500 g/l – Cat. A/j: two-component high-performance coating (BS – solvent-based)
Tool cleaning	UNI solvent
Storage	12 months, in a dry place at a temperature between 5°C and 30°C; pay particular attention to component B, which is sensitive to air humidity

APPLICATION DATA AND TIMES

DESCRIPTION	RESULT
Mixing ratio (by weight)	A = 100, B = 33
Pot life (50% RH)	at 10°C: 70 min; at 25°C: 40 min; at 35°C: 25 min
Touch dry (50% RH)	at 10°C: 20-24 hours; at 25°C: 12-14 hours; at 35°C: 8-10 hours
Foot traffic (50% RH)	at 25°C: 12-24 hours

Overcoating (50% RH)	at 25°C: from 12 to 36 hours (after sanding)
Full traffic (50% RH)	at 25°C: 3-5 days
Full curing (50% RH)	at 25°C: 7 days
Ambient conditions for use	Temperature between +10°C and +35°C, RH < 60%. DS-POLTEC SOL must be applied at a substrate temperature at least 3°C above the dew point, to avoid adhesion problems.
Maintenance of the coating	Wait at least 7 days after application of the coating before carrying out cleaning operations

TECHNICAL PERFORMANCE DATA

DESCRIPTION	RESULT
Appearance	Smooth and uniform
Gloss at 60°	30-40 (*)
Gloss at 85° (DIN 67530)	40-70 (*)
Slip resistance (Italian Ministerial Decree 236/89)	μ > 0,40 (with 3% FINE GLASS BEADS)
Slip resistance (DIN 51130)	R11 (with 5% COARSE GLASS BEADS); R12 (with 20% QUARTZ B2)
CE marking (Reg. No. 305/2011)	Compliant with EN 1504-2. Coating (C), Principle (PR)
Pull-off adhesion (EN 1542)	> 1,5 MPa
Abrasion resistance (EN ISO 5470-1)	< 50 mg (Taber CS-10, 1000 cycles, 1000 g)
BCA wear resistance (EN 13892-4)	AR 0,5
Impact resistance (EN ISO 6272-1)	14 N · m
Water permeability (EN 1062-3)	w < 0,1 kg/(m^2 · h^0,5)

(*) The gloss level of the applied DS-POLTEC SOL is influenced by the porosity of the substrate and the thickness of the film.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.